

SAI VIDYUT CHANDRAMOHAN

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PROFESSIONAL SUMMARY

Second-year Computer Science student who builds ambitious end-to-end software and AI systems — turning ambiguous problems into working products across backend architecture, applied AI, and full-stack engineering.

EXPERIENCE

Product & Backend Intern — Studojo Labs Private Limited

May 2026 – Jul 2026

- Contributed to the design and development of backend services supporting core product workstreams over a two-month internship, collaborating with the engineering team from design through implementation.
- Owned assigned backend deliverables end-to-end within a live software engineering team environment.

SELECTED PROJECTS

Project Blueprint — AI Software Planning & Architecture Tool

- Built an AI-powered planning tool that converts a natural-language product idea into a developer-grade blueprint — project summary, MVP scope, architecture, tech-stack rationale, database schema, API design, auth strategy, deployment plan, and roadmap.
- Architected a strict separation between probabilistic and deterministic layers: the AI produces structured, schema-validated (Zod) architecture output, which the application then deterministically compiles into Mermaid diagrams and roadmap visualizations, rather than having the model emit diagram syntax directly.
- Designed a resilient multi-provider AI layer (Gemini, Cerebras, Groq, Hugging Face, OpenRouter) with automatic fallback across unavailable providers, on a staged-generation Next.js/TypeScript app with live progress tracking.

Technologies: Next.js, TypeScript, Tailwind CSS, shadcn/ui, Zod, multi-provider AI integration

SatQuery — AI Satellite Intelligence Platform

- Architected an end-to-end platform (FastAPI, Next.js, MapLibre) that turns natural-language questions about Earth-surface change into a planned investigation, using a deterministic planner with an optional LLM-based planner to select the relevant analysis tools.
- Built multimodal remote-sensing analysis on Google Earth Engine — Sentinel-2 optical change detection (CVA, cloud masking, vectorized regions), Sentinel-1 SAR backscatter change detection, and Dynamic World semantic layers — fused into a single evidence-backed result.
- Integrated a standalone GeoChat remote-sensing VLM as a GPU inference service for scene captioning and bi-temporal, cross-modal optical+SAR change understanding, validated with automated backend and frontend E2E tests.

Technologies: Python, FastAPI, Next.js, MapLibre, Google Earth Engine, Sentinel-1/2, Pydantic

RazorFlow — Agentic Commerce on Razorpay

- Designed an agentic commerce system that converts conversational buyer intent into governed transactions: AI-driven intent understanding and product ranking feed into deterministic merchant policy guardrails before any transaction executes.
- Built the full commerce loop — role-authenticated buyer/staff/admin flows, Razorpay checkout with server-side signature verification, payment webhooks, and recovery flows — on Next.js/TypeScript, PostgreSQL, and Prisma, with unit/integration/E2E test coverage.

Technologies: Next.js, TypeScript, PostgreSQL, Prisma, Razorpay APIs

Project DocNA — Document Intelligence Pipeline

- Built a format-agnostic document intelligence pipeline that separates AI reasoning from document mutation: hybrid structural + LLM detection generates semantic answers, a deterministic placement policy converts them into structured operations, and an adapter performs controlled OOXML edits via lxml/zipfile on a copy — always preserving the immutable original.
- Validated the pipeline with 311 automated tests and 18/18 passing real-AI evaluation fixtures.

Technologies: Python 3.12+, FastAPI, Pydantic v2, python-docx, lxml

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++

Backend & APIs: FastAPI, REST API design, Pydantic, SQLAlchemy

AI/LLM: Structured AI generation, multi-provider AI integration, agentic workflow design, prompt engineering, remote-sensing VLM integration

Frontend: React, Next.js, Tailwind CSS, MapLibre

Data & Geospatial: PostgreSQL, MySQL, Prisma, Google Earth Engine, Sentinel-1/2 imagery

Tools & Infra: Git/GitHub, Docker, Azure Blob Storage, OOXML/lxml, Ollama

EDUCATION

B.Tech, Computer Science and Engineering — SRM Institute of Science and Technology (SRMIST)

2nd Year

ACHIEVEMENTS

- First Place — Li-Fi (Light Fidelity) light-based data transmission system (semiconductor engineering project); Gold from Garbage Initiative (recycled-materials engineering model); Engineering Model Development coursework.